Hierarchical Projection as a Validity Layer for Multilingual ESG Promise Verification

Sheng-Xun Chen, Chun-Wei Chen, Hsin-Chen Han, Shi-Yu Huang

National Central University, Taiwan • Team_10537

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1 Introduction

- ESG promise verification labels each paragraph with **four linked fields**, and the output is meant for audit-oriented use.
- A record is only usable if its fields agree, yet independent classifiers can combine plausible labels into an impossible tuple.
- The official metric scores each field on its own, so an invalid tuple still earns **44.3%** of the weighted field credit.

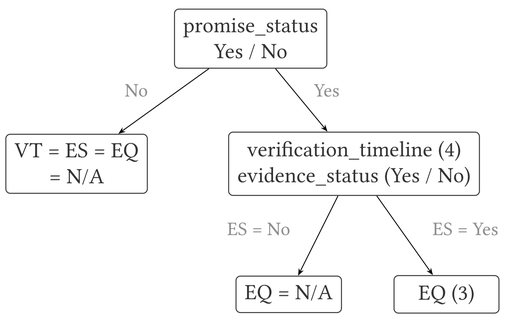
Research question. *Can a retraining-free decision rule guarantee legal outputs, and how does it change field-wise versus whole-tuple scores?*

2 Task & Data

- Fields: promise status (PS), verification timeline (VT), evidence status (ES), evidence quality (EQ).
- Hard rules: PS = No $\Rightarrow$ VT = ES = EQ = N/A; ES = No $\Rightarrow$ EQ = N/A.
- Only **17 of 120** label combinations are legal.
- **AI CUP 2026**: 2,000 Chinese paragraphs from 49 reports, all legal (15 of 17 states observed).
- **ML-Promise**: English, French, Japanese (400 paragraphs each), Korean (500 PDF pages).

3 Methodology

(a) Task hierarchy



$1 + 4 \times (1 + 3) = 17$ legal tuples

of $2 \times 5 \times 3 \times 4 = 120$ combinations

the other 103 are invalid

(b) Decision rules: output rule × calibration

base probabilities (identical for every method)

calibration

	none	global	conditional
independent argmax *may be invalid*	**M0**	—	—
projection *always valid*	**M1**	**M2**	**M3**
17-state decoding *always valid*	**M4**	**M5**	**M6**

the seven rules are alternatives, not stages: every cell consumes the same base probabilities

Figure 1: the label hierarchy and the seven alternative decision rules, all reading the same base probabilities

Validity-preserving output rules

- **Projection**: greedy and top-down. Pick PS; under PS = Yes take the best substantive VT and ES; under ES = Yes the best substantive EQ. Always legal, no retraining.
- **17-state decoding**: return the legal state with the highest summed log-score.
- Calibration: none, global, or conditional class biases.

Evaluation

- `hfl/chinese-roberta-wwm-ext-large`, four independent heads.
- Five-way cross-fitting, three seeds, document-disjoint split.
- Paired report-cluster bootstrap, Holm-adjusted.
- External: 4 backbones $\times$ $\lambda \in \{0, 0.3\}$ in each of 4 languages = **32 arms**.

4 Results: Chinese Anchor

ID	Calibration	Decoding	PS	VT	ES	EQ	wF1 (official)	Tuple acc.	Invalid %
M0	None	Independent	0.796	0.464	0.650	0.424	0.572±0.003	0.359±0.007	12.55
M1	None	Projection	0.796	0.467	0.651	0.418	0.571±0.003	0.394±0.004	0.00
M2	Global	Projection	0.796	0.493	0.650	0.418	0.574±0.004	0.428±0.004	0.00
M3	Conditional	Projection	0.796	0.501	0.651	0.416	0.575±0.003	0.432±0.004	0.00
M4	None	17-state	0.799	0.468	0.648	0.420	0.571±0.005	0.388±0.003	0.00
M5	Global	17-state	0.796	0.493	0.649	0.423	0.576±0.001	0.430±0.001	0.00
M6	Conditional	17-state	0.784	0.494	0.634	0.418	0.567±0.008	0.427±0.005	0.00

Table 2: three-seed means ± s.d., document-disjoint

Metric	Contrast	Δ	95% CI	p_{Holm}
wF1 (official)	M1-M0	-0.001	[-0.006, 0.003]	1.000
	M3-M2	+0.001	[-0.004, 0.006]	1.000
	M4-M1	+0.000	[-0.005, 0.005]	1.000
	M6-M3	-0.008	[-0.018, 0.002]	0.416
	M6-M5	-0.008	[-0.016, -0.001]	0.171
Tuple accuracy	M1-M0	+0.035	[0.028, 0.043]	**0.001**
	M3-M2	+0.004	[-0.001, 0.008]	0.370
	M4-M1	-0.006	[-0.010, -0.002]	**0.028**
	M6-M3	-0.005	[-0.015, 0.004]	0.505
	M6-M5	-0.004	[-0.011, 0.004]	0.505

Table 4: pre-specified paired contrasts, 95% intervals

- Projection removes every invalid tuple (**12.55% → 0**) and raises tuple accuracy by **+0.035** ($p_{\text{Holm}} = .001$), while official wF1 does not move ($\Delta = -0.001$, $p_{\text{Holm}} = 1.000$).
- Searching all 17 states is not better: joint decoding loses **−0.006** tuple accuracy to projection ($p_{\text{Holm}} = .028$).

5 Multilingual Replication

Corpus	Rows	M0 invalid %	N/A net	Subst. net	Δ wF1	Δ tuple acc.
Chinese (AI CUP)	2,000	12.55	+395	-264	-0.0011	**+0.0350**
English	400	23.17	+148	-86	+0.0032	**+0.0725**
French	400	21.08	+150	-107	+0.0036	**+0.0625**
Japanese	400	31.25	+183	-245	-0.0131	**+0.0708**
Korean	500	22.73	+103	-148	-0.0192	**+0.0293**

Table 7: projection (M1) vs. argmax (M0), one preselected arm per corpus

- Tuple accuracy rises in **32/32** external arms; wF1 in only **14/32** (EN 7/8, FR 5/8, JA 0/8, KO 2/8).
- These are within-arm directions, not a ranking of languages.

6 Why the Metrics Disagree

- A wrong parent makes projection overwrite a correct child with N/A; a right parent lets it repair an unsupported child.
- N/A credit goes up and substantive credit goes down in all five corpora, yet whole tuples improve in every one.
- Joint decoding revises PS on **339 of 12,000** rows: 225 become correct and 114 wrong, but whole-row correctness on them falls **114 → 47**.
- Fixing a parent rarely saves a row whose children stay wrong; breaking a correct parent always loses it.

7 Implications for Audit Use

- Validity is a **precondition** for a usable record, not proof that the record is right.
- A repair can hide an inconsistency, so a deployed validity layer should **log the rows it repaired**.
- Pick the metric by how outputs are used:
  - records: tuple accuracy among valid outputs
  - single-field actions: that field's F1, or rare-class recall
  - portfolio statistics: class-level scores, keeping validity
  - leaderboards: official wF1, but not for gaps under 0.008

8 Conclusion

- Hierarchical projection is a **retraining-free validity layer**: it eliminates invalid tuples by construction.
- It improves whole-tuple accuracy on the Chinese anchor and in all 32 external arms.
- Its field-score effect varies across models and corpora; its logical guarantee does not.

Takeaway. For audit use, make validity a precondition, and pick the metric that matches the unit at which errors have consequences.

References

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